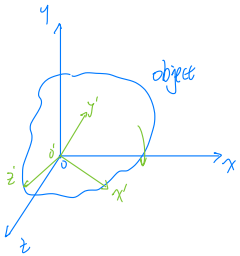
· Transport Equation



typically the origin of moving frame B is chosen to be the center of mass

frame I: $\{O, x, y, z\}$

frame B: $\{O', x', y', z'\}$

let V be a vector

V_I be the description of V in I

V_B be the description of V in B

$$V_I = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \end{bmatrix} V_B$$

let $\frac{d}{dt} V_I$ be the change of V in I frame

$\frac{d}{dt} V_B$ be the change of V in B frame

$$\frac{d}{dt} V_I = \frac{d}{dt}\left(\begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \end{bmatrix} V_B \right)$$

$$= \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \end{bmatrix} \frac{d}{dt} V_B + \begin{bmatrix} | & | & | \\ w \times \hat{i} & w \times \hat{j} & w \times \hat{k} \\ | & | & | \end{bmatrix} V_B$$

$$= \left(\frac{d}{dt} V_B\right)_I + [w_I] \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \end{bmatrix} V_B$$

$$= \underbrace{\left(\frac{d}{dt} V_B\right)_I}_{\text{velocity of } V \text{ in } B \text{ described in } I} + \underbrace{w_I \times V_I}_{\text{rotation of } B \text{ wrt } I}$$

$$\frac{d}{dt}(\cdot)_I = \left(\frac{d}{dt}(\cdot)_B\right)_I + w_I \times (\cdot)_I$$

$$\frac{d^2}{dt^2} V_I = \frac{d}{dt}\frac{d}{dt} V_I = \frac{d}{dt}\left[\left(\frac{d}{dt} V_B\right)_I + w_I \times V_I\right]$$

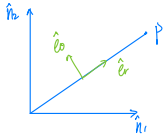
$$= \frac{d}{dt}\left(\frac{d}{dt} V_B\right)_I + \frac{d}{dt}(w_I \times V_I)$$

$$= \left(\frac{d}{dt}\left(\frac{d}{dt} V_B\right)_B\right)_I + w_I \times \left(\frac{d}{dt} V_B\right)_I + \dot{w}_I \times V_I + w_I \times \frac{d}{dt} V_I$$

$$= \left(\frac{d^2}{dt^2} V_B\right)_I + w_I \times \left(\frac{d}{dt} V_B\right)_I + \dot{w}_I \times V_I + w_I \times \left[\left(\frac{d}{dt} V_B\right)_I + w_I \times V_I\right]$$

$$= \left(\frac{d^2}{dt^2} V_B\right)_I + \underbrace{2 \times w_I \times \left(\frac{d}{dt} V_B\right)_I}_{\text{Coriolis acc}} + \underbrace{w_I \times w_I \times V_I}_{\text{Centrifugal w/r}} + \underbrace{\dot{w}_I \times V_I}_{\text{tangential acc due to } \dot{w}}$$

- Polar Coordinate



Inertial frame $I = \{O, \hat{n}_1, \hat{n}_2\}$

polar frame $B = \{O, \hat{e}_r, \hat{e}_\theta\}$

$$ {}^{I}r_{P/O} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} {}^{B}r_{P/O} $$

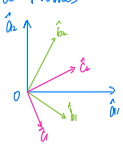
$$ \frac{d}{dt}(r_{P/O})_{/I} = \frac{d}{dt}(r\cdot\hat{e}_r)_{/I} $$
$$ = \dot{r}\cdot\hat{e}_r + r\cdot\frac{d}{dt}(\hat{e}_r)_{/I} $$
$$ = \dot{r}\cdot\hat{e}_r + r\cdot\left[\left(\frac{d}{dt}(\hat{e}_r)_{/B}\right)_{/I} + \omega_{I/I}\times\hat{e}_{r/I}\right] $$
$$ = \dot{r}\cdot\hat{e}_r + r\dot{\theta}\hat{e}_\theta $$

$$ \frac{d}{dt}(r_{P/O})_{/I} = \dot{r}\cdot\hat{e}_r + r\dot{\theta}\hat{e}_\theta \quad \text{Inertial velocity} $$

$$ \frac{d^2}{dt^2}(r_{P/O})_{/I} = \frac{d}{dt}\left(\dot{r}\cdot\hat{e}_r + r\dot{\theta}\hat{e}_\theta\right) $$
$$ = \ddot{r}\hat{e}_r + \dot{r}\frac{d}{dt}\hat{e}_r + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\frac{d}{dt}\hat{e}_\theta $$
$$ = \ddot{r}\hat{e}_r + \dot{r}\left[\left(\frac{d}{dt}\hat{e}_{r/B}\right)_{/I} + \omega_{I/I}\times\hat{e}_{r/I}\right] + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\left[\left(\frac{d}{dt}\hat{e}_{\theta/B}\right)_{/I} + \omega_{I/I}\times\hat{e}_\theta\right] $$
$$ = \ddot{r}\hat{e}_r + \underbrace{2\dot{r}\dot{\theta}\hat{e}_\theta}_{\text{coriolis}} + \underbrace{r\ddot{\theta}\hat{e}_\theta}_{\text{tangential}} - \underbrace{r\dot{\theta}^2\hat{e}_r}_{\text{centrifugal}} $$

$$ \frac{d}{dt}(r_{P/O})_{/I} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{e}_\theta $$

- Multiple Frames



$A = \{O, \hat{a}_1, \hat{a}_2\}$ Inertial Frame

$B = \{O, \hat{b}_1, \hat{b}_2\}$

$C = \{O, \hat{c}_1, \hat{c}_2\}$

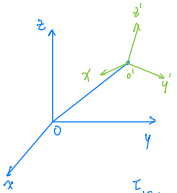
$$ \frac{d}{dt}V_{/B} = \left(\frac{d}{dt}V_{/C}\right)_{/B} + \omega_{B/B}\times V_{/B} $$

$$ \frac{d}{dt}V_{/A} = \left(\frac{d}{dt}V_{/B}\right)_{/A} + \omega_{B/A}\times V_{/A} $$
$$ = T_{AB}\left[\left(\frac{d}{dt}V_{/C}\right)_{/B} + \omega_{B/B}\times V_{/B}\right] + \omega_{B/A}\times V_{/A} $$
$$ = \left(\frac{d}{dt}V_{/C}\right)_{/A} + \omega_{B/A}\times V_{/A} + \omega_{B/A}\times V_{/A} $$
$$ = \left(\frac{d}{dt}V_{/C}\right)_{/A} + (\omega_{B/A} + \omega_{B/A})\times V_{/A} $$

as if C is rotating about $\omega_{B/A} + \omega_{B/A}$

rotation axis can be added

Rotating & Translating Frame



(typically the origin of moving frame (A) is chosen to be the center of Mass

$$\vec{r}_{P/O} = \vec{r}_{O'/O} + \vec{r}_{P/O'}$$

$$\Downarrow \frac{d^2}{dt^2}$$

$$\frac{{}^I d^2}{dt^2} r_{P/O} = \frac{{}^I d^2}{dt^2} r_{O'/O} + \frac{{}^I d^2}{dt^2} r_{P/O'}$$

$$\vec{F}_p = m \frac{{}^I d^2}{dt^2} r_{O'/O} + m \frac{{}^I d^2}{dt^2} r_{P/O'}$$

fictious force due to acc of B wrt I

Eg: pendulum in accelerating box

